
MICROSOFT POWER BI PART I

INTRODUCTION

POWER BI OVERVIEW

- Power BI is a technology-driven business intelligence tool provided by Microsoft for analyzing and visualizing raw data to present actionable information.
 - It combines business analytics, data visualization, and best practices that help an organization to make data-driven decisions.
 - Microsoft as Leader in the
- "2024 Gartner Magic Quadrant for **Analytics and Business Intelligence Platforms**"



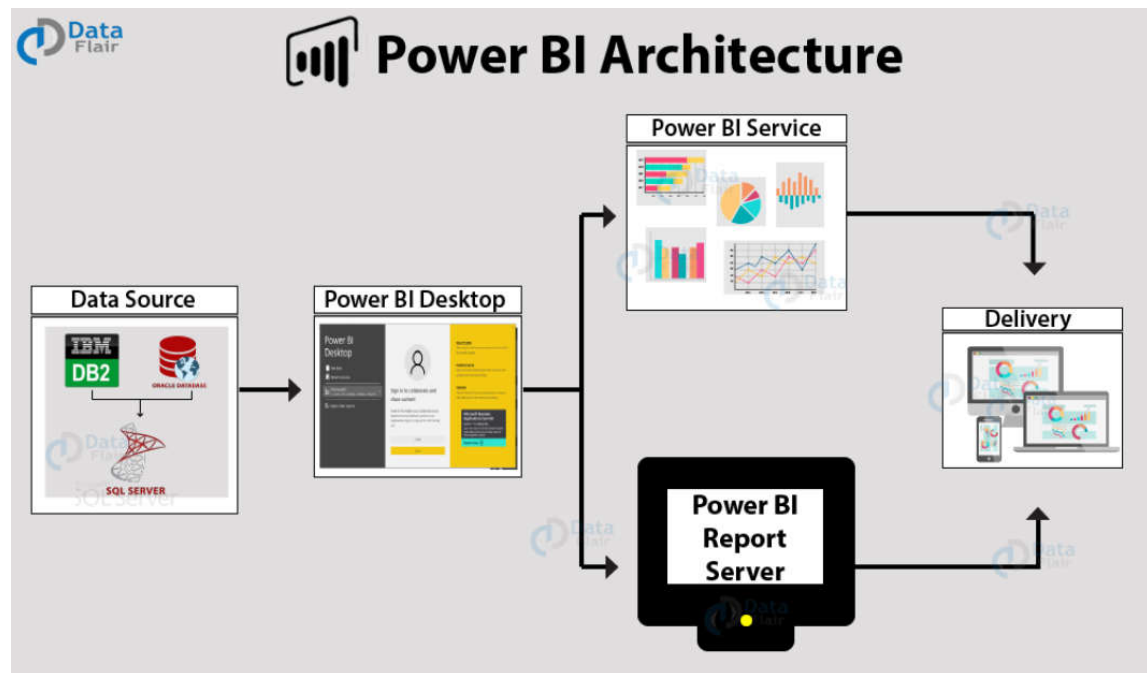
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POWER BI ARCHITECTURE

- Power BI gives the ability to analyze and explore data on premise as well as in the cloud.
- Power BI architecture is a service built on top of Azure. There are multiple data sources that Power BI can connect to.



POWER BI ARCHITECTURE

- **Power BI desktop** is a component primarily used for Business reports generation and desktop creation. Also, all the other works typically begin with Power BI desktop, where the report creation takes place.
- **Power BI services** refer to the cloud services that are used to publish Power BI reports and data visualizations. Power BI service is the Software as a Service (SaaS) part of Power BI.
- Using **Power BI mobile apps**, you can stay connected to their data from anywhere. Power BI apps are available for Windows, iOS, and Android platforms.

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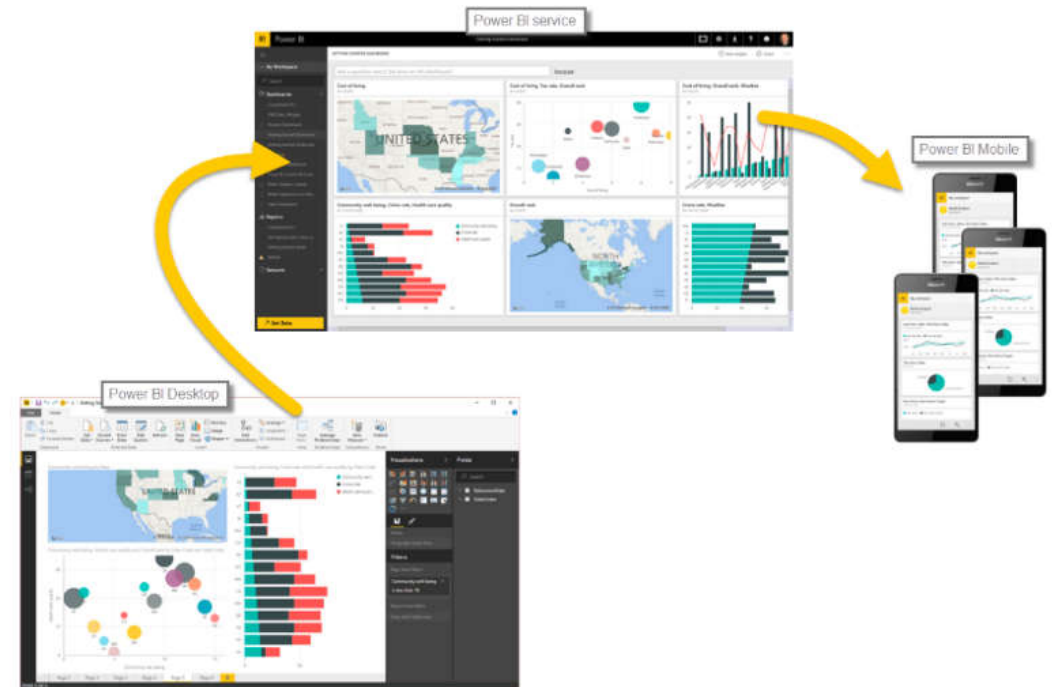
POWER BI OVERVIEW

- Power BI provides the ability to collaborate and share customized dashboards and interactive reports across colleagues and organizations, easily and securely.



POWER BI DESKTOP

- Usually, an analysis work begins in **Power BI Desktop** where report creation takes place.
- The report is then published to **Power BI service** from where it can be shared to the **Power BI Mobile apps** so that people can view the reports even on mobiles.



POWER BI DESKTOP

- Power BI Desktop is a free application that can be downloaded and installed on the system.
- The most common uses for Power BI Desktop are as follows:
 - Connect to multiple data sources
 - Transform and clean that data, to create a data model
 - Create visuals, such as charts or graphs, that provide visual representations of the data
 - Create reports that are collections of visuals, on one or more report pages
 - Share reports with others by using the Power BI service

POWER BI DESKTOP WORKSPACE

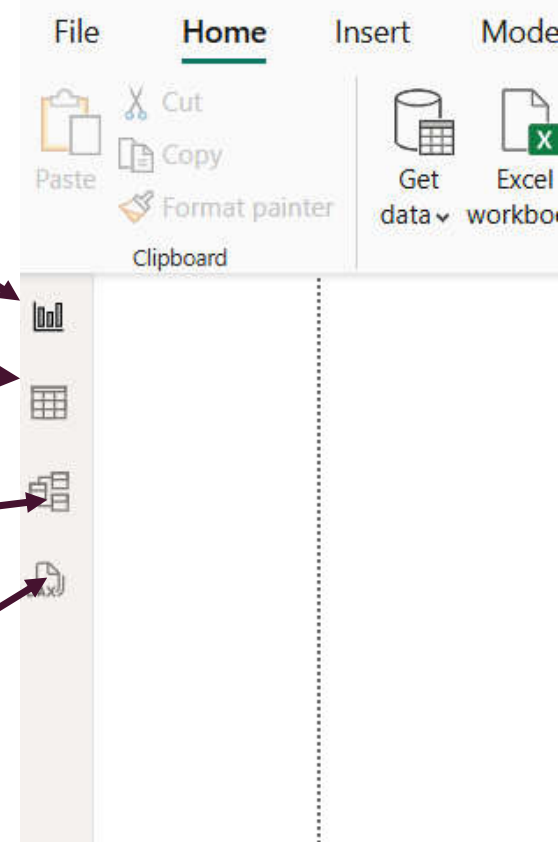
■ Basic Views

- **-Report View:** In this view, you create reports and visuals, where most of your creation time is spent.

-Table View: The table view gives a preview of the entire data. Tables, measures, and other data used in the data model associated with the report, and transform the data for best use in the report's model

-Model View: The model view displays the relationship between various objects.

- **-DAX query view-** view and work with Data Analysis Expressions (DAX) queries on semantic models.



REPORT VIEW

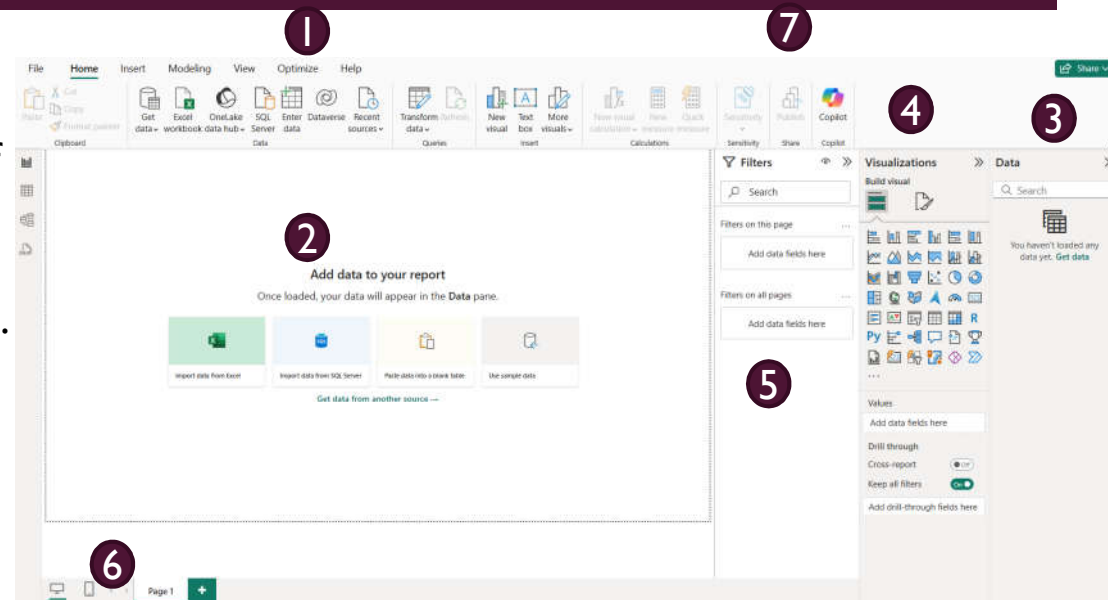
1. Ribbon - the top ribbon contains most of the controls and options needed for building the report.

- **Get Data:** allows you to import your data in the form of excel files, text/csv, xml, etc. We can also obtain data from Microsoft Azure, or other online services like google analytics, GitHub, etc.
- 2. **Canvas** : where visualizations are created and arranged.
- 3. **Datas**: contains the tables and fields that are available in the data model.
- 4. **Visualizations**: contains the list of available visualizations.
- **Fields** - this section contains the tables and fields that are available in the data model.

5. **The Filters pane**: where you can filter data visualizations

6. **The pages tab** :for navigation to other pages in the report.

7. **Publish**: enables you to publish your data model to your Power BI workspace.



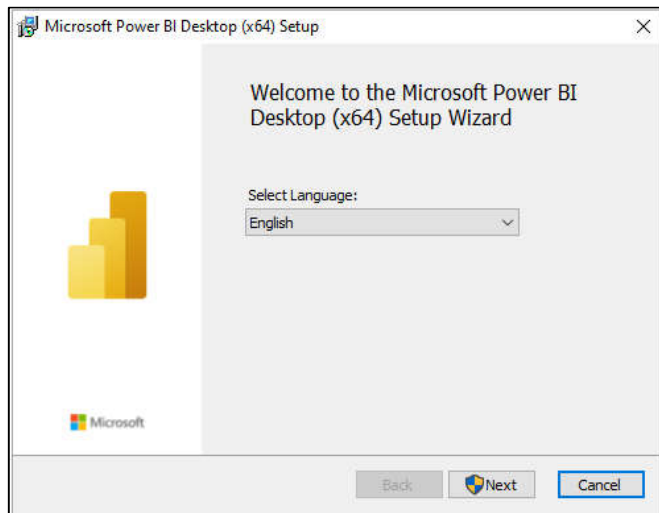
REPORTS

- A collection of visuals, in one Power BI Desktop file, is called a report.
- A report can have one or more pages, just like an Excel file can have one or more worksheets.
- With Power BI Desktop you can create complex and visually rich reports, using data from multiple sources, all in one report that you can share with others in your organization.



INSTALLATION

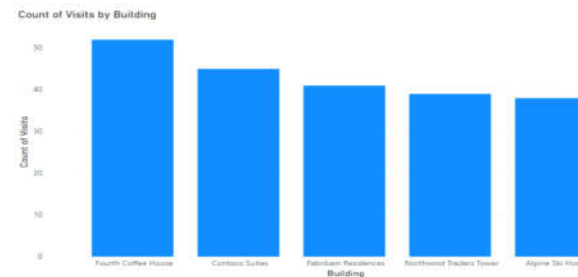
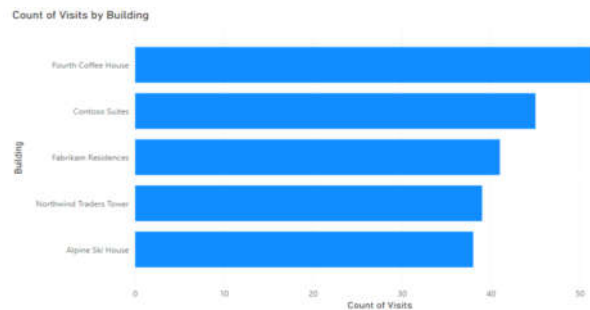
- To download the Power BI Desktop executable from the Download Center, select **Download** from the Download Center page (<https://www.microsoft.com/download/details.aspx?id=58494>).
- Follow through the installation wizard until finish.



TYPES OF VISUALISATIONS

1. Bar and column charts

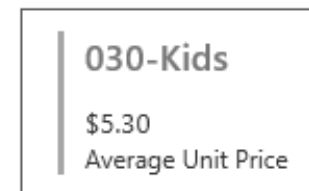
Bar charts are the standard for looking at a specific value across different categories.



2. Cards: Single number



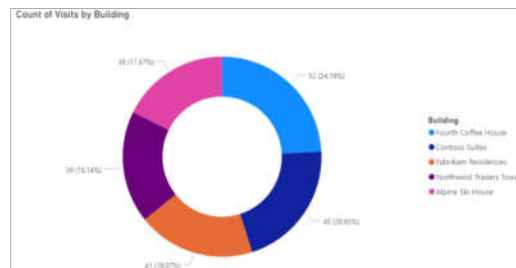
3. Cards: Multi row



TYPES OF VISUALISATIONS

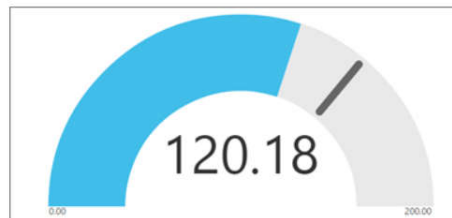
4. Donut charts

Donut charts are similar to pie charts. They show the relationship of parts to a whole.



6. Gauge charts

Displays current status in the context of a goal.



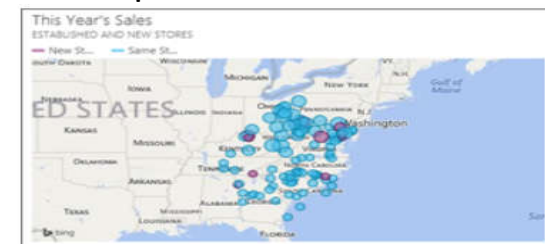
5. Pie charts

Pie charts show the relationship of parts to a whole.



7. Maps: Basic maps

Used to associate both categorical and quantitative information with spatial locations.



TYPES OF VISUALISATIONS

8. Matrix

A table supports two dimensions, but a matrix makes it easier to display data meaningfully across multiple dimensions -- it supports a stepped layout. The matrix automatically aggregates the data and enables drill down.

Quarter	Q1	Q2
Year	Revenue	YTD Revenue
2010	\$45,186,241.942528	45,186,241.94
2011	\$47,641,801.21002861	47,641,801.21
2012	\$53,054,664.287495166	53,054,664.29
2013	\$45,011,474.33002325	45,011,474.33
2014	\$45,648,089.42003315	45,648,089.42
2015	\$45,186,241.942538336	45,186,241.94
Total	\$281,728,513.13264656	45,186,241.94

10. Q&A visual

The Q&A visual lets you ask questions about your data using natural language.

Try one of these to get started

top geo states by total units

top geo states by total OTHER units

top geo states by total compete volume

Show all suggestions

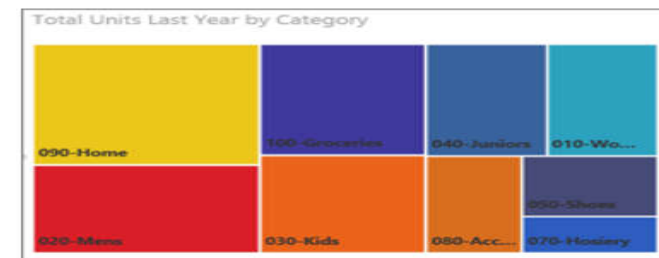
9. Tables

Works well with quantitative comparisons among items where there are many categories.

District	Count of StoreNumber	Average Selling Area Size
FD - 01	9	43,889
FD - 02	9	47,778
FD - 03	9	50,000
FD - 04	10	50,500
LI - 01	13	10,385
LI - 02	11	10,909
Total	61	33,361

11. Treemaps

Are charts of colored rectangles, with size representing value. They can be hierarchical, with rectangles nested within the main rectangles.

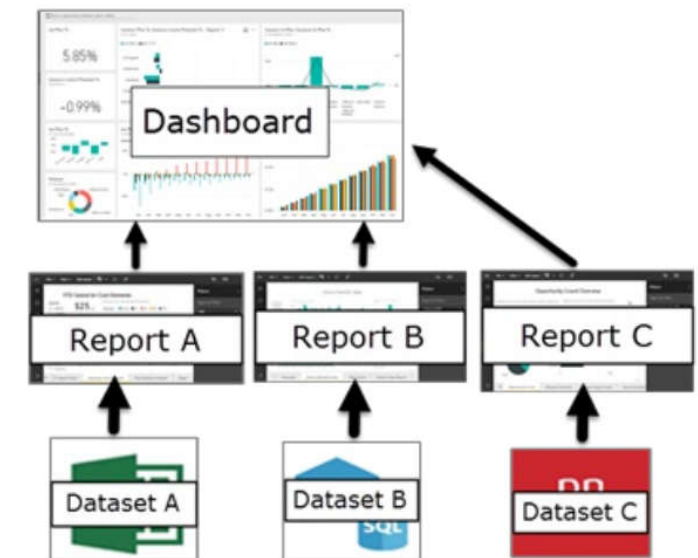


POWER BI SERVICE

- Power BI is a collection of software services, apps, and connectors that work together to create, share, and consume business insights in the way that serves you and your business most effectively.
- The Microsoft Power BI service (app.powerbi.com), sometimes referred to as Power BI online, is the SaaS (Software as a Service) part of Power BI.
- In the Power BI service, dashboards help to keep a finger on the pulse of the business. Dashboards display tiles, which user can select to open reports for exploring further.
- Dashboards and reports connect to datasets that bring all of the relevant data together in one place

DASHBOARD

- A Power BI dashboard is a single page, often called a canvas, that tells a story through visualizations. Because it's limited to one page, a well-designed dashboard contains only the highlights of that story. Readers can view related reports for the details.
- Dashboards are a feature of the Power BI service only. They're not available in Power BI Desktop. Although you can't create dashboards on mobile devices, you can view and share them there
- The visualizations you see on the dashboard are called tiles. You pin tiles to a dashboard from reports.
- The visualizations on a dashboard originate from reports and each report is based on a dataset. Selecting a visualization takes you to the report (and dataset) that it's based on



DATA SET

- The dataset you will use today is a sales and market share analysis. This type of analysis is very common for the office of a Chief Marketing Officer (CMO). Unlike the office of the Chief Financial Officer (CFO), a CMO is focused not only on company's performance internally (how well do our products sell) but also externally (how well do we do against the competing products).
- The company, VanArsdel, manufactures expensive electronic products that could be used for fun as well as work and it sells them directly to consumers nationwide as well as several other countries. VanArsdel and its competitors have retained a 3rd party marketing company to collect and anonymize industry sales so that all participants can benchmark themselves.
- Download the dataset (USSales and Internatioal sales and copy all files in folder name DIAD/Data



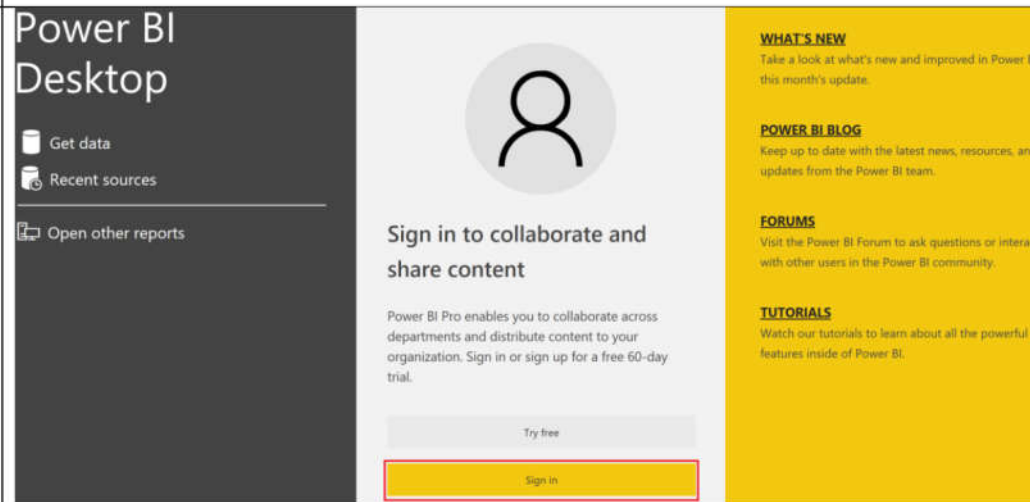
MICROSOFT POWER BI

DATA PREPARATION

POWER BI DESKTOP - LOAD DATA

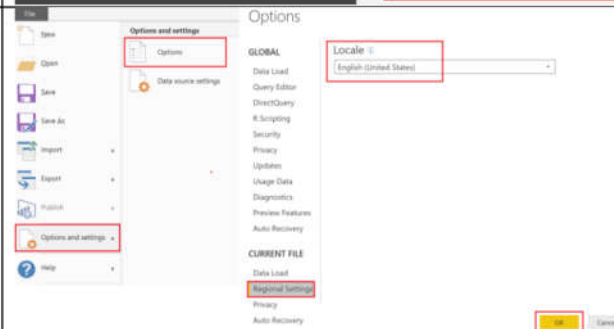
Steps

1. If you don't have the **Power BI Desktop** open, launch it now.
2. Startup screen opens. Click on **Sign In** and sign in using your Power BI credentials. Signing in to Power BI Desktop helps later when you are publishing to Power BI Service.



Let's set up the locale to US English, to make it convenient to go through the rest of this lab.

3. From the ribbon, select **File -> Options and settings -> Options**.
4. In the left panel of Options dialog, select **Regional Settings**.
5. From the **Locale** drop down select **English (United States)**.



POWER BI DESKTOP - LOAD DATA

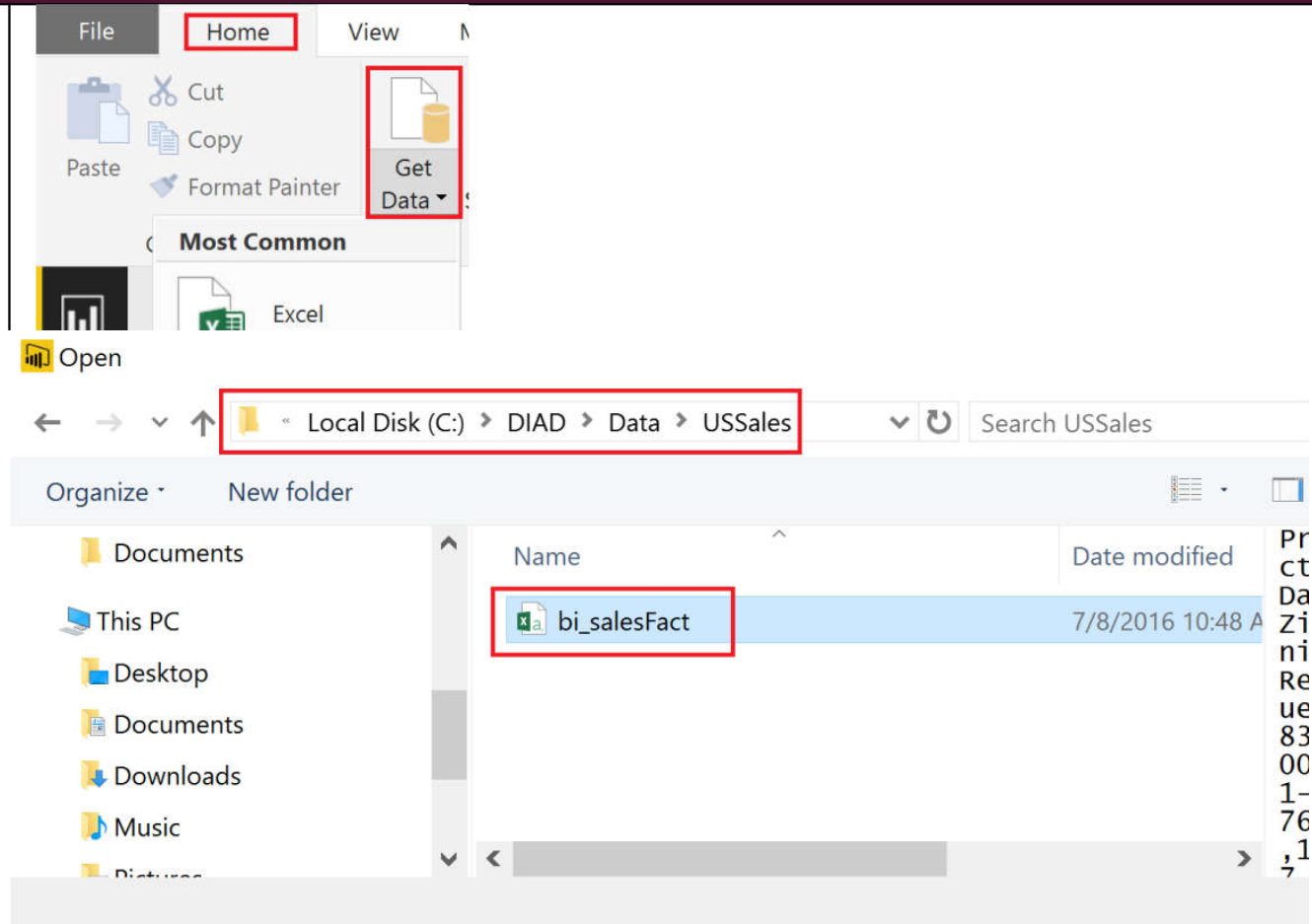
First step is to [load data](#) to Power BI Desktop. We will load USA Sales data which is in CSV files.

6. From the ribbon, select **Home** -> **Get Data**.

7. Select **Text/CSV**.

8. Browse to **DIAD\Data\USSales** folder and select **bi_salesFact.csv**.

9. Click **Open**.



POWER BI DESKTOP - LOAD DATA

- After completing your selection, you have three options – Load, Transform data or Cancel.
- Click Transform data.
- Now you have loaded the Sales data into the file. You should be in the Query Editor window as shown in the picture to the right. Notice data type of each field is indicated next to the column header

bi_salesFact.csv

File Origin: 1252: Western European (Windows) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

ProductID	Date	Zip	Units	Revenue
2359	19/9/2003	41083	1	430.4475
976	19/9/2003	32822	1	314.9475
2379	19/9/2003	37873	1	178.4475
767	18/7/2001	26257	1	104.9475
2254	18/7/2001	34428	1	57.6975
962	19/7/2001	97002	1	524.9475
2074	19/7/2001	6762	1	330.6975
725	19/7/2001	1475	1	115.4475
796	27/9/2001	95650	1	278.1975
443	27/9/2001	55350	1	814.8
2359	26/9/2001	94954	1	440.9475
740	26/9/2001	95215	1	173.1975
2284	26/9/2001	33411	1	299.1975
709	27/9/2001	94534	1	262.4475
2047	27/9/2001	76301	1	388.4475
396	27/9/2001	52677	1	876.4875
740	27/9/2001	93444	1	173.1975
2250	27/9/2001	89074	1	125.9475
740	27/9/2001	75965	1	173.1975
1916	27/9/2001	64801	1	280.8225

The data in the preview has been truncated due to size limits.

Extract Table Using Examples | Load | Transform Data | Cancel

Untitled - Power Query Editor

File | Home | Transform | Add Column | View | Tools | Help

Close & Apply | New Source | Recent Sources | Enter Data | Data source settings | Manage Parameters | Refresh Preview | Properties | Advanced Editor | Manage | Choose Columns | Remove Columns | Keep Rows | Remove Rows | Sort | Split Column | Group By | Data Type: Whole Number | Merge Queries | Append Queries | Combine Queries

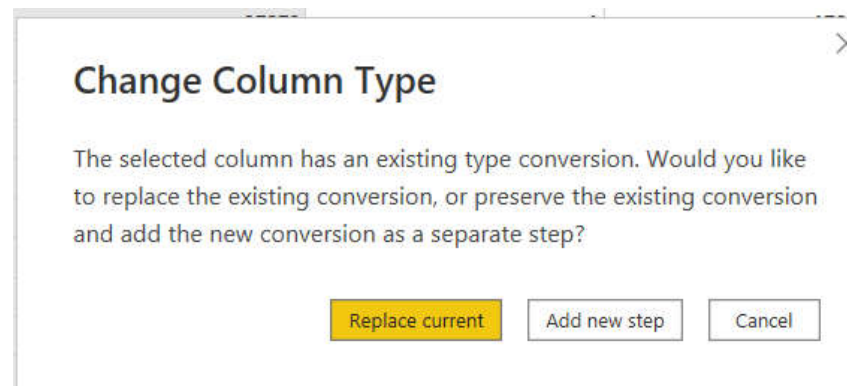
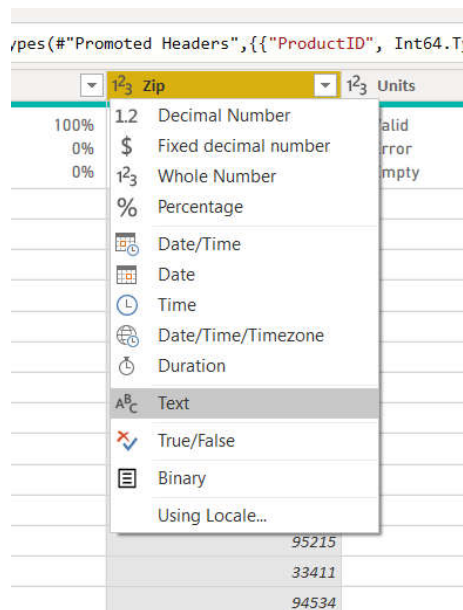
Queries [1] | bi_salesFact

Table.TransformColumnTypes(#"Promoted Headers",{{"ProductID", Int64.Type}, {"Date", type text}, {"Zip", Int64.T...})

	ProductID	Date	Zip	Units	Revenue
1	2359	9/19/2003	41083	1	430.4475
2	976	9/19/2003	32822	1	314.9475
3	2379	9/19/2003	37873	1	178.4475
4	767	7/18/2001	26257	1	104.9475
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6	962	7/19/2001	97002	1	524.9475
7	2074	7/19/2001	6762	1	330.6975
8	725	7/19/2001	1475	1	115.4475
9	796	9/27/2001	95650	1	278.1975
10	443	9/27/2001	55350	1	814.8

POWER BI DESKTOP - LOAD DATA

- Notice Power BI has set Zip field to data type Whole Number. To ensure that Zip codes which start with zero don't lose the leading zero, we will format them as text. Highlight the Zip column and change the Data Type to Text.
- Change Column Type dialog opens. Select Replace Current button.



POWER BI DESKTOP - LOAD DATA

- Now let's get the data that is in excel source file.
- 1. From the ribbon, select Home -> New Source -> Excel
- 2. Browse to DIAD\Data\USSales folder and select bi_dimensions.xlsx.
- 3. In the Navigator window select all the tables. Select each table to preview the columns and rows. Note: The first four items are the named Excel Tables and the second set are the Excel Worksheet names. Table names are differentiated from Worksheet names by using different icons.
- 4. Click OK to edit these tables in the query editor. Notice 4 new queries are added to the query editor.

Navigator

The screenshot shows the Power BI Navigator window. On the left, under 'Display Options', the file 'bi_dimensions.xlsx [8]' is listed. Below it, four tables are selected with checkboxes: 'bi_date', 'bi_geo', 'bi_manufacturer', and 'bi_product'. These are highlighted with a red box. Below these are four worksheet names: 'date', 'geo', 'manufacturer', and 'product', which are not selected. On the right, a preview of the 'bi_product' table is shown. It has columns: ProductID, Product, Category, Segment, and ManufacturerID. The table contains 14 rows of data. At the bottom right, there are 'OK' and 'Cancel' buttons, with the 'OK' button highlighted by a red box.

ProductID	Product	Category	Segment	ManufacturerID
1	Abbas MA-01	Mix	All Season	1
2	Abbas MA-02	Mix	All Season	1
3	Abbas MA-03	Mix	All Season	1
4	Abbas MA-04	Mix	All Season	1
5	Abbas MA-05	Mix	All Season	1
6	Abbas MA-06	Mix	All Season	1
7	Abbas MA-07	Mix	All Season	1
8	Abbas MA-08	Mix	All Season	1
9	Abbas MA-09	Mix	All Season	1
10	Abbas MA-10	Mix	All Season	1
11	Abbas MA-11	Mix	All Season	1
12	Abbas MA-12	Mix	All Season	1
13	Abbas MA-13	Mix	All Season	1
14	Abbas MA-14	Mix	All Season	1

POWER BI DESKTOP - LOAD DATA

- To analyze all the data together you will want to import the new data from each of the subsidiaries and combine it with the US Sales you loaded earlier.
- 1. Click on the New Source drop down in the Home menu tab of the Query Editor.
- 2. Select More... as shown in the figure. Get Data dialog opens.. In the Get Data dialog select Folder. Click Connect
- 3. Folder dialog opens. Click Browse... button. In the Browse for Folder dialog navigate to the InternationalSales folder.
- 4. Click OK (to close the Browse for Folder dialog box) and Click OK (to close the Folder dialog box).
- 5. Since we want to combine data, click Combine & Transform Data

Content	Name	Extension	Date accessed	Date modified	Date created	Attributes
Binary	AU Sales.csv	.csv	24/3/2026 10:20:21 PM	26/7/2018 9:44:22 PM	25/6/2025 9:10:48 AM	Record
Binary	CA Sales.csv	.csv	24/3/2026 10:20:12 PM	26/7/2018 9:44:22 PM	25/6/2025 9:10:49 AM	Record
Binary	FR Sales.csv	.csv	24/3/2026 10:20:11 PM	26/7/2018 9:44:22 PM	25/6/2025 9:10:49 AM	Record
Binary	JP Sales.csv	.csv	24/3/2026 10:20:04 PM	26/7/2018 9:44:22 PM	25/6/2025 9:10:49 AM	Record
Binary	MX Sales.csv	.csv	24/3/2026 10:19:43 PM	26/7/2018 9:44:22 PM	25/6/2025 9:10:50 AM	Record

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Combine & Transform Data

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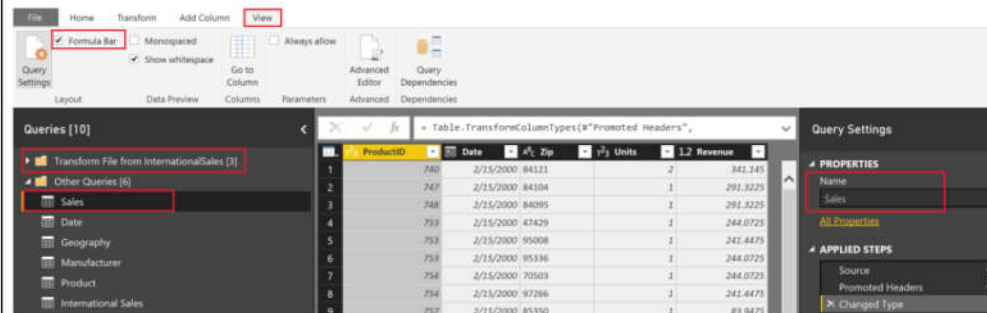
POWER BI DESKTOP - TRANSFORMING YOUR DATA

The Query Editor window should appear as shown in the diagram.

- If formula bar is disabled, you can turn on the formula bar from the View ribbon. This enables you to see the “M” code generated by each click on the ribbons.
- Select the options available on the ribbon – **Home, Transform, Add Column and View** to notice the various features available.

1. Under **Queries** panel, **minimize** Transform Binary from InternationalSales folder.
2. Select each query name in the **Other Queries** section.
3. **Rename** them in the Query Settings -> Properties section as shown below:

Initial Name	Final Name
bi_salesFact	Sales
bi_date	Date
bi_geo	Geography
bi_manufacturer	Manufacturer
bi_product	Product
InternationalSales	International Sales



4. From the left panel, select **Date** Query.
5. Using Ctrl left click key select **Year, Quarter, MonthNo, MonthName and MonthID** fields.
6. From the ribbon select **Transform -> Fill -> Down**.

Notice now all the null values are filled with the appropriate Year, Quarter, MonthNo, MonthName, MonthID values.

Queries [10]

- Transform File from InternationalSales [3]
- Other Queries [6]
 - Sales
 - Date**
 - Geography
 - Manufacturer
 - Product
 - International Sales

Table:

	Year	Quarter	MonthNo	MonthName	MonthID	Date
1	1999	Q1	1	Jan	199901	1/1/1999
2	null	null	null	null	null	1/2/1999
3	null	null	null	null	null	1/3/1999
4	null	null	null	null	null	1/4/1999
5	null	null	null	null	null	1/5/1999
6	null	null	null	null	null	1/6/1999
7	null	null	null	null	null	1/7/1999
8	null	null	null	null	null	1/8/1999
9	null	null	null	null	null	1/9/1999

Formulas: = Table.TransformColumnTypes(bi_date_Table,{{"Year", Int64.Type}, {"Quarter", type

Ribbon: Transform

- Group By
- Use First Row As Headers
- Count Rows
- Table
- Data Type: Any
- Detect Data Type
- Rename
- Fill
- Down
- Up
- Column
- Convert To List
- Unpivot Columns
- Move
- Split Column
- Format
- Parse
- Merge Columns
- Extract
- Text Column
- Statistics
- Standard
- Scientific
- Trigonometry
- Rounding
- Information
- Number Column
- Date

Formulas: = Table.FillDown("#Changed Type",{"Year", "Quarter", "MonthNo", "MonthName", "Mon

Queries [10]

- Transform File from InternationalSales [3]
- Other Queries [6]
 - Sales
 - Date**
 - Geography
 - Manufacturer
 - Product
 - International Sales

Table:

	Year	Quarter	MonthNo	MonthName	MonthID	Date
1	1999	Q1	1	Jan	199901	1/1/1999
2	1999	Q1	1	Jan	199901	1/2/1999
3	1999	Q1	1	Jan	199901	1/3/1999
4	1999	Q1	1	Jan	199901	1/4/1999
5	1999	Q1	1	Jan	199901	1/5/1999
6	1999	Q1	1	Jan	199901	1/6/1999
7	1999	Q1	1	Jan	199901	1/7/1999
8	1999	Q1	1	Jan	199901	1/8/1999
9	1999	Q1	1	Jan	199901	1/9/1999

7. From the left panel, select **Manufacturer** Query. Notice ManufacturerID and Manufacturer data is laid across in rows. And the header is not useful. We need to transpose the table to meet our needs.

8. From the ribbon select **Transform -> Transpose**.

Notice this transposes the data into columns. Now we need the first row to be the header.

9. From the ribbon select **Home -> Use First Row As Headers**.

Notice now Manufacturer table is laid out the way we need it with a header and values along columns.

Also, notice in the **APPLIED STEPS** section on the right panel, all the steps that we add is being recorded.

The screenshot shows the Power Query Editor interface. The 'Transform' ribbon is active, and the 'Transpose' button is highlighted with a red box. The 'Queries [10]' pane on the left shows the 'Manufacturer' query selected, also highlighted with a red box. The main area displays the formula bar with the query name and a preview of the data table. The table has two columns: 'ManufacturerID' and 'Manufacturer'. The data is as follows:

	ManufacturerID	Manufacturer
1	1	2
2	Abbas	Aliqui
	Barba	Currus

10. From the left panel, select **Product** Query.
- Notice Product field is a concatenation of Product Name and Product ID. For our scenario, we need only the Product Name.
11. Select **Product** column.
12. From the ribbon select **Home** -> **Split Column** -> **By Delimiter**.
13. Dialog opens. From the dialog, select - **Custom**—from the dropdown.
14. Enter “-” in the text area since we want to split the column based on the hyphen.
15. Select **OK**.

Notice Product column is split into two columns, Product.1 and Product.2.

We do not need Product.2 column since it is the Product ID, which is already available.

16. Select **Product.2** column.
17. From the ribbon select **Home** -> **Remove Columns** to remove the column.

Let's rename Product.1 column to Product.

18. Right click on **Product.1** column header.
19. From the menu select **Rename** to rename the column to **Product**.

The first screenshot shows the 'Split Column by Delimiter' dialog box. The 'Select or enter delimiter' dropdown is set to 'Custom--' and the text area contains a hyphen '-'. The 'Split at' options are 'each occurrence of the delimiter'.

The second screenshot shows the 'Remove Columns' ribbon tab. The 'Product.2' column is selected in the column list on the left, and the 'Remove Columns' button is highlighted in the ribbon.

The third screenshot shows the 'Rename' option in the context menu. The 'Product.1' column header is right-clicked, and the 'Rename' option is selected from the menu.

20. From the left panel, select **Sales** Query.

21. Verify and if needed, **Replace Current** Data Type using **Home -> Data Type** drop down as shown below:

Column	Data Type
Date	Date
Revenue	Fixed Decimal Number

Select **Replace Current** in the Change Column Type dialog.

The screenshot shows the Power BI Desktop interface. The 'Home' tab is selected in the ribbon. The 'Data Type' dropdown menu is open, showing various data types. The 'Fixed decimal number' option is highlighted. The 'Queries' panel on the left shows the 'Sales' query selected. The main view displays a table with columns: ProductID, Date, Zip, Units, and Revenue. The 'Revenue' column is highlighted in yellow.

	ProductID	Date	Zip	Units	Revenue
1	740	2/15/2000	84121	2	341.145
2	747	2/15/2000	84104	1	291.3225
3	748	2/15/2000	84095	1	291.3225
4	753	2/15/2000	47429	1	244.0725
5	753	2/15/2000	95008	1	241.4475
6	753	2/15/2000	95336	1	244.0725
7	754	2/15/2000	70503	1	244.0725
8	754	2/15/2000	97266	1	241.4475
9	757	2/15/2000	85350	1	83.9475

22. From the left panel, select **Date** Query.

23. Verify and if needed **Replace Current** Data Type using **Home** -> **Data Type** drop down as shown below.

Column	Data Type
Date	Date

24. From the left panel, select **Manufacturer** Query.

25. Verify and if needed set Data Type using **Home** -> **Data Type** drop down as shown below.

Column	Data Type
ManufacturerID	Whole Number
ManufacturerName	Text

The screenshot shows the Power Query Editor interface. The 'Home' tab is selected in the ribbon. The left panel shows the 'Queries [10]' list, with the 'Date' query selected. The main area displays a table with columns: Year, Quarter, MonthNo, MonthName, MonthID, and Date. The 'Date' column has a dropdown menu open, showing 'Data Type: Date'. A text box overlay states: 'IMPORTANT! Changing the data type is a big deal to use later'.

	Year	Quarter	MonthNo	MonthName	MonthID	Date
1	1999	Q1	1	Jan	199901	1/1/1999
2	1999	Q1	2	Jan	199901	1/2/1999
3	1999	Q1	3	Jan	199901	1/3/1999
4	1999	Q1	4	Jan	199901	1/4/1999
5	1999	Q1	5	Jan	199901	1/5/1999
6	1999	Q1	6	Jan	199901	1/6/1999
7	1999	Q1	7	Jan	199901	1/7/1999
8	1999	Q1	8	Jan	199901	1/8/1999
9	1999	Q1	9	Jan	199901	1/9/1999

26. From the left panel, select **International Sales Query**.

27. Verify and if needed set Data Type using **Home -> Data Type** drop down as shown below.

Column	Data Type
ProductID	Whole Number
Date	Date
Zip	Text
Units	Whole Number
Revenue	Fixed Decimal Number

The screenshot shows the Power BI Desktop interface. The 'Home' tab is selected in the ribbon. The 'Data Type' dropdown menu is open, showing various data types. The 'Fixed decimal number' option is highlighted. The 'Queries' list on the left shows 'International Sales' selected. The main view displays a table with columns: ProductID, Date, Zip, Units, and Revenue. The 'Revenue' column is highlighted in yellow.

Queries [10]

- Transform File from InternationalSales [3]
- Other Queries [6]
 - Sales
 - Date
 - Geography
 - Manufacturer
 - Product
 - International Sales**

Table Data:

	ProductID	Date	Zip	Units	Revenue	
1	2331	2/14/2002	3042	1	540.6975	Aust
2	733	7/10/2004	1022	1	330.6975	Aust
3	753	7/10/2004	838	1	246.6975	Aust
4	964	7/10/2004	1158	1	320.1975	Aust
5	400	7/10/2004	854	1	772.275	Aust
6	499	7/10/2004	854	1	911.6625	Aust
7	683	2/8/2002	9009	1	839.7375	Aust
8	714	2/8/2002	1117	1	162.6975	Australia
9	1105	2/8/2002	7907	1	246.6975	Australia

Data Type: Fixed decimal number

- Decimal Number
- Fixed decimal number**
- Whole Number
- Percentage
- Date/Time
- Date
- Time
- Date/Time/Timezone
- Duration
- Text
- True/False
- Binary

Formula Bar: = Table.TransformColumnTypes(#"Removed Columns",{{"Revenue", Curre

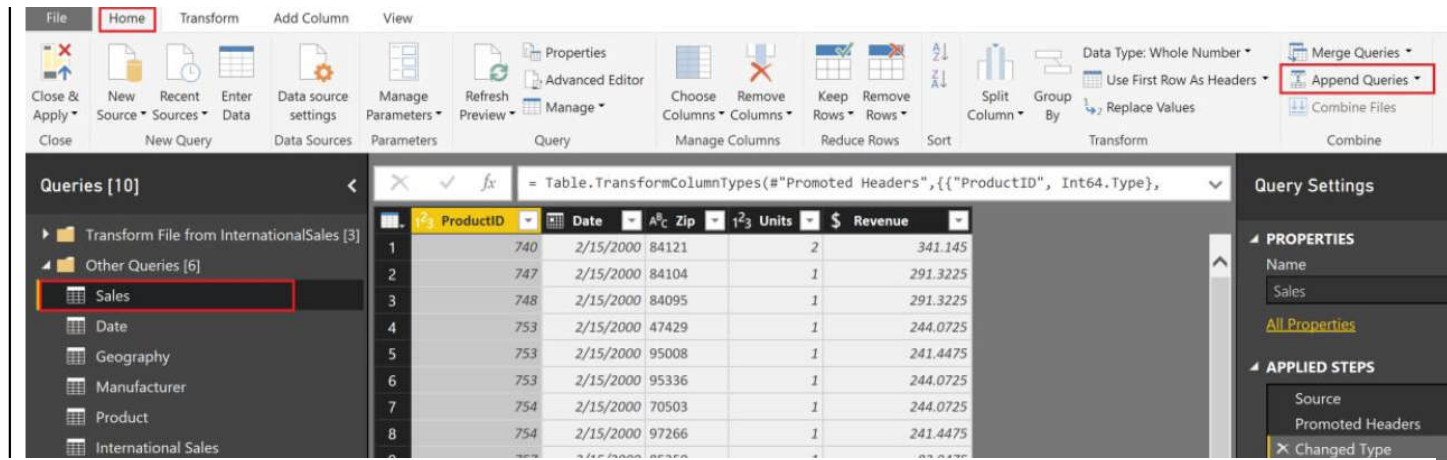
To analyze the Sales of all countries, it is convenient to have a single Sales table. Hence you want to append all the rows from **International Sales** to **Sales**.

28. Select **Sales** in the Queries window in the left panel as shown in the figure.

29. From the ribbon select **Home -> Append Queries** as shown in the figure.

Append dialog opens. There is an option to append **Two tables or Three or more tables**. Leave Two tables selected since we are appending just two tables.

30. Select **International Sales** from the drop down and click **OK**.



Append

☒ Two tables ☐ Three or more tables

Table to append

International Sales

OK

Cancel

You will now see a new column in the **Sales** table called **Country**. Since International Sales had the additional column for Country, Power BI Desktop added the column to the Sales table when it loaded the values from International Sales.

You see **null values** in the **Country** column by default for the Sales table rows because the column did not exist for the table with USA data.

We will add the value “**USA**” as a data shaping operation.

31. From the ribbon select **Add Column -> Conditional Column** as shown in the figure.

The screenshot shows the Power BI Desktop interface. The 'Add Column' ribbon is active, with the 'Conditional Column' option highlighted. The 'Queries' pane on the left shows the 'Sales' query selected. The main view displays a table with the following data:

	ProductID	Date	Zip	Units	Revenue	Country
1	740	2/15/2000	84121	2	341.145	null
2	747	2/15/2000	84104	1	291.3225	null
3	748	2/15/2000	84095	1	291.3225	null
4	753	2/15/2000	47429	1	244.0725	null
5	753	2/15/2000	95008	1	241.4475	null
6	753	2/15/2000	95336	1	244.0725	null
7	754	2/15/2000	70503	1	244.0725	null
8	754	2/15/2000	97266	1	241.4475	null
9	757	2/15/2000	85350	1	83.9475	null

32. In the **Add Conditional Column** dialog, enter name of the column as “**CountryName**”.
33. Select **Country** from the **Column Name** dropdown.
34. Select **equals** from the **Operator** dropdown.
35. Enter **null** in the **Values** text.
36. Enter **USA** in the **Output** text.
37. Select the dropdown under **Else** and pick **Select a column** option.
38. Select **Country** from the column dropdown.
39. Click **OK**.

×

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

CountryName

Column Name

Operator

Value ①

Output ①

If

Country

equals

ABC 123

null

Then

ABC 123

USA

...

Add Clause

Else ①

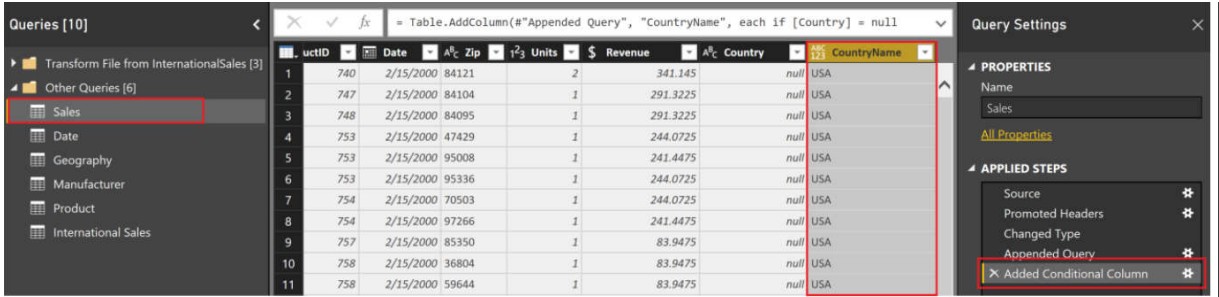
Country

OK

Cancel



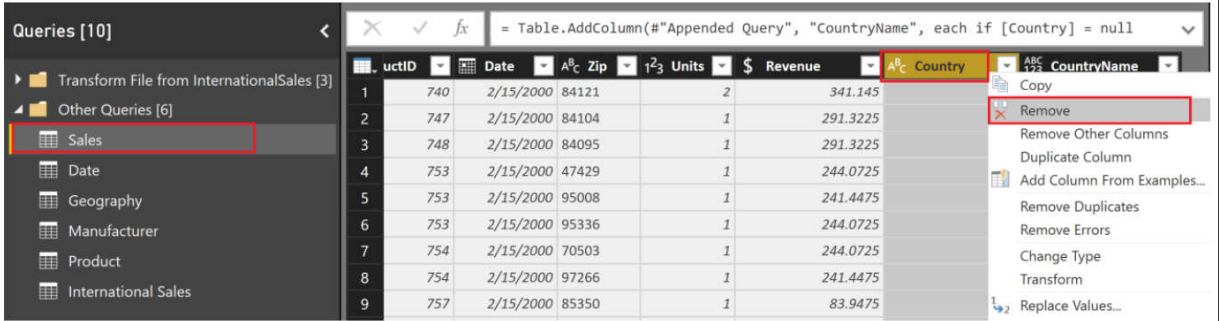
40. You will see the **CountryName** column in the Query editor window.



The original **Country** column is only required as a temporary column. It is not required in the final table for analysis and can be removed.

41. Right click on the **Country** column and select **Remove** as shown in the figure.

We can now rename **CountryName** column to **Country**.



42. Right click on the **CountryName** column and rename to **Country**.

43. Using **Home -> Data Type**, change the **data type** of the **Country** column to type **Text**.

The newly named **Country** column will have names for all countries, including the USA. You can validate this by clicking on the drop down next to **Country** column to see the unique values.

44. At first, you will only see USA data. Click on **Load more** to validate you have data from all 4 countries included.

45. Click **OK** to close this filter.

The screenshot shows the Power BI Desktop interface. On the left, the 'Queries' pane lists 'Sales' as the selected query. The main view displays a table with columns: ProductID, Date, Zip, Units, Revenue, and Country. A filter dropdown is open for the 'Country' column, showing a list of countries with checkboxes: (Select All), Australia, Canada, France, Japan, Mexico, and USA. The 'OK' button is highlighted in the bottom right corner of the dropdown menu.

ProductID	Date	Zip	Units	Revenue	Country
740	2/15/2000	8412			USA
747	2/15/2000	8410			USA
748	2/15/2000	8409			USA
753	2/15/2000	4742			USA
753	2/15/2000	9500			USA
753	2/15/2000	9533			USA
754	2/15/2000	7050			USA
754	2/15/2000	9726			USA
757	2/15/2000	8535			USA
758	2/15/2000	3680			USA
758	2/15/2000	5964			USA
767	2/15/2000	3387			USA
767	2/15/2000	9311			USA
768	2/15/2000	6484			USA
768	2/15/2000	9226			USA
769	2/15/2000	0809			USA
769	2/15/2000	3428			USA
797	2/15/2000	7501			USA

- We don't need the International Sales table to load to the data model now that its rows have been appended to the Sales table. Let's prevent the International Sales table from loading to the data model.
- From the Queries panel on the left, select International Sales query.
- Right click and select Enable Load. This will disable loading International Sales to prevent duplicate data.
- Select Home – Close and Apply

